INCH-POUND

A-A-55699
31 March 1995

COMMERCIAL ITEM DESCRIPTION

PATIENT MONITOR

The General Services Administration has authorized the use of this Commercial Item Description.

This Commercial Item Description covers a patient monitor for use in vital signs monitoring indoors, outdoors, and during transport.

Salient Characteristics:

The unit shall be a small, lightweight, ultra-portable, battery and line powered patient monitor, capable of monitoring patient ECG, heart rate, invasive blood pressure, non-invasive blood pressure, pulse rate, and temperature; displaying selected parameters on a LCD screen and strip chart recorder. The monitor shall have maximum flexibility to allow utilization as follows:

a. Individual bedside monitoring with capability to display at a central location.

b. Patient monitoring during emergency transport.

c. Patient monitoring in the operating room during the administration of anesthesia.

d. Patient monitoring after surgery, during transport to, and during recovery.

e. Patient monitoring during transport to and while doing special procedures.

f. Patient monitoring in hostile environments during transport while exposed to the outdoors.

AMSC N/A FSC 6515

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The unit shall be capable of providing an ECG through a shielded, three lead patient cable having a six pin connector, configured in accordance with the latest edition of the AAMI ECG Standard for ECG Connectors.

The front panel of the unit shall be provided with a pushbutton lead selection switch for leads I, II, and III.

The monitor display shall use liquid crystal (LCD) dot matrix technology of the super twist type.

The unit shall have a user selectable sweep speed of 12.5, 25, and 50mm per second and an amplitude sensitivity of 10mm per millivolt (mV).

Common Mode Rejection Ratio (CMRR) shall be less than 10mV with respect to the input. Measurement to be 20V at 60 Hz with 51K ohms of misbalance in lead source resistance.

The frequency response of the ECG signal amplifier shall not be more than 3dB down from 0.5 to 40 Hz. The display frequency response shall be at least 0.5 Hz at 50mm/second.

The noise level shall be less than 40mVpp RTI in any position of the lead select switch.

A one centimeter reference pulse shall be provided so that if the trace is deflected off scale due to lead handling or lead selection, it shall return to within 1mm of its original position within 4.0 seconds.

An automatic trace reset shall be provided so that if the trace is deflected off scale due to lead handling or lead selection, it shall return to within 1mm of its original position within 4.0 seconds.

The unit shall withstand a defibrillator energy pulse of 360 joules into a 50 ohm load parallel to the ECG leads.

The ECG input impedance shall exceed 2.5 Megohms for frequencies up to 10 Hz. The unit shall operate with common mode input offsets of up to 300mV.

The monitor shall be capable of displaying heart rate with an input from ECG electrodes or pulse rate from the non-invasive blood pressure.

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There shall be an internally computed and digitally displayed heart rate with a range of 25 to 250 beats per minute. Shall count biphasic 0.2mV to 5mV "R" waves with no external sensitivity adjustment.

The digital display of the value of heart rates shall be updated successively at intervals not to exceed 4 seconds.

There shall be a digital display capable of selectively providing the following physiological data:

a. Heart rate in beats per minute derived from ECG or pulse rate signal.

b. Temperature in degrees Centigrade or Fahrenheit.

c. Blood pressure in mmHg: systolic, diastolic, mean pressure.

The monitor shall have an audible and visible alarm indicator for heart rate, pulse rates, systolic, diastolic, and mean blood pressure which violate chosen high and low limits. The audible alarm shall be able to be manually reset. A temporary alarm silence shall last no more than two minutes.

The high and low alarm limits shall be independently adjustable and read on the LCD, with a setting accuracy of ±1 BPM, 1mm Hg, 1°C. All alarm parameters and limits should have the capability of being set with one button activation.

The ECG monitor shall show pacemaker pulses on the ECG waveform display, but shall not count the pacemaker pulses on the ECG waveform display, nor shall not count the pacemaker pulse in computing the heart rate.

Battery operation. To allow for monitoring during power failures and emergency transport, the unit shall be capable of operating from a battery pack that is to be included with the unit.

a. Battery type: Rechargeable, sealed lead acid.

b. Battery capacity: 6.5 hours of continuous monitoring or 1 hour of continuous recording.

c. Battery display: Shall indicate voltage of batteries.

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d. Battery charge indicator: Shall illuminate when batteries are charging.

e. Battery charge time: 8 hours maximum to full charge for each depleted battery.

Batteries will have at least a 5 year shelf-life with no greater than 30 percent permanent loss in capacity if stored at 20°C.

Blood pressure. Shall contain two (2) invasive and one (1) non-invasive channels. The output shall be such that the pressures can be displayed on a direct writing recorder and simultaneous waveform displays on the LCD as well as processed to digital display of systolic, diastolic, and mean pressure.

a. Invasive channels: The range for the invasive blood pressure channels shall be from -30 to 300mm Hg for systolic, diastolic, and mean blood pressure. Pulse range shall be from 25 to 250 BPM. An auto zero feature or a zeroing control shall be provided to compensate for any residual output voltage that may exist without any applied pressure. If a zero balance control is furnished, it shall allow adjustment over a range of ±200mm Hg for 5.0 VDC excitation voltage.

b. Non-Invasive channel: Systolic range: 30 to 270mm Hg. Diastolic range: 20 to 250mm Hg. Mean pressure range: 25 to 260mm Hg. Pulse range 25 to 250 BPM. Shall indicate a pressure display accuracy of ±3mm Hg or 2 percent whichever is greater. Response time for determining blood pressure shall be between 28 to 45 seconds with a maximum determination time of 3 minutes. Maximum cuff pressure shall be 275mm Hg with cuff over-pressure relief valve at 370mm Hg. Child, standard adult, large adult, and thigh blood pressure cuffs shall be provided.

Temperature.

a. Input: The unit shall include a sensor suitable for rectal, skin, or esophageal temperature measurement. Unit shall be compatible with Yellow Springs Instrument Co., 400 and 700 series probes and Electromedics 2100, 2400, and 2700 series probes.

b. Range: The temperature measuring function (including probe) shall cover the range of 17°C to 50°c (62°F to 122°F).

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c. Accuracy: The temperature range shall be digitally displayed with an accuracy of ±0.2°C (0.3°F).

Recorder. The instrument shall be able to accept an expansion module such that the unit can add a printer and remain as one functional portable unit. A direct writing recorder shall be furnished for intermittent recording of up to three (3) continuous waveforms. The recorder shall be capable of automatically recording the direct or delayed waveform signals when manually initiated or when an alarm condition occurs.

a. A standard paper speed of 25mm per second shall be provided with an accuracy of ±5 percent.

b. The recording paper shall be of the thermal type and available from a number of commercial sources. The paper recording output shall be suitable for inclusion in the patient's chart. The paper shall allow ball-point pen and pencil writing. The paper recording shall also provide clear copies after copying in a copying machine. The recording paper shall be ruled with 5.0mm divisions along both the time and voltage axis. The 5.0mm divisions shall not deviate more than ±2 percent from 5.0mm in either axis. The ruled division shall span, along the voltage axis, at least 4cm. The recordings shall be rectilinear.

c. Drift of the recording due to all causes shall not exceed 1.0mm of vertical displacement per 10mm of horizontal deflection.

d. The band width of the recorder should be 0.5 to 40 Hz.

e. The triggering of an alarm will automatically initiate the direct paper charting of the alarmed patient's ECG. The recording shall contain a minimum of 12 seconds of information at 20mm/second.

f. The user shall be able to initiate a strip chart recording of the trace information displayed on the associated waveform display device. The trace may continue for a fixed time not to exceed 20 seconds.

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Shall be provided with but not limited to the following accessories:

a. Two (2) patient cables (10 feet (3.03 meter) minimum length).

b. Three (3) pre-gelled electrodes.

c. Two (2) spare batteries.

d. Three (3) packages recording paper.

The unit power cord assembly, shall be approximately 6 feet long with a hospital grade plug.

Shall meet the electrical characteristics as stated below:

The electrical characteristics of the monitor shall be as follows:

115 ±10 percent, 50 ±2 Hz
115 ±10 percent, 60 ±2 Hz
230 ±10 percent, 50 ±2 Hz
230 ±10 percent, 60 ±2 Hz

Operational and Service Data Manuals shall be furnished as follows:

a. Information content:

The contractor shall furnish manuals, handbooks, and/or brochures containing complete operation, installation, and service/maintenance instructions (including pictures of illustrations, as necessary) with complete schematics and wiring diagrams. All manuals, handbooks, and/or brochures will be written in the English language. All schematics brochures will be written in the English language, and all schematics and wiring diagrams will use American electrical and electronics symbols. The manuals will include electrical data and connection diagrams for all applicable utilities. The instructions shall also contain a complete list of all replaceable parts showing part number, name, and quantity required. All service notes, service memos, etc., pertaining to the equipment and issued subsequent to the printing of the original instructions shall also be included. These instructions shall be, as a minimum, those furnished to service engineers (or servicemen) who normally install and

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service the equipment for the company of distributor. When the system being procured includes, or will operate ancillary equipment, the service data shall include complete instructions and drawings which show interfacing of all system components.

b. Quantities, distribution and schedules:

Two (2) complete sets of the above information shall be packaged with each unit of the required equipment. Where multiple quantities of the same unit are to be delivered to the same ultimate consignee, then only a total of two (2) service manuals needs to be supplied to each consignee. However, two (2) operation manuals still need to be supplied with each unit.

Workmanship. The patient monitor shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One patient monitor, as specified, constitutes one unit. Patient monitor shall be supplied in a suitable unit carton.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

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Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, with an Acceptable Quality Level (AQL) of 1.0 (percent defective) and an inspection level of S-2. The unit of product for examination purposes shall be one patient monitor.

Sampling for tests. Sampling for tests, including dimensional test, shall be conducted in accordance with MIL-STD-105, with an AQL of 1.0 (percent defective) and an inspection level of S-2. The unit of product for test purposes shall be one patient monitor.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

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Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order.

NOTE: The following National Stock Number(s) is/are covered by this document.

National Stock Number	Item Identification
6515-01-391-7008	Patient Monitor

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MILITARY INTERESTS:

Custodians:

Air Force - 03
Army - MD
Navy - MS

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

FDA-MPQAS
USPHS
VA-OSS

Project Number: 6515-5751

Location: PMON17C.DOC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1 The preparing activity must complete blocks 1, 2, 3 and 8 In block 1, both the document number and revision letter should be given

2 The submitter of this form must complete blocks 4 5, 6, and 7

3 The preparing activity must provide a reply within 30 days from receipt of the form

NOTE This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements

I RECOMMEND A CHANGE:	1 DOCUMENT NUMBER A-A-55699	2 DOCUMENT DATE (YYMMDD) 950331

3 DOCUMENT TITLE

PATIENT MONITOR

4 NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible Attach extra sheets as needed)

5 REASON FOR RECOMMENDATION

6 SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8 PREPARING ACTIVITY

a NAME DEFENSE PERSONNEL SUPPORT CENTER ATTN: DPSC-MAP	b TELEPHONE (Include Area Code) (1) Commercial (215)737-8431	(2) AUTOVON 444-8431
c ADDRESS (Include Zip Code) 2800 South 20th Street Philadelphia, PA 19145	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756 2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 Previous editions are obsolete

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